Abstract

Objectives: Quantitative spinal cord MRI biomarkers exhibit systematic vendor-dependent variability that limits multi-centre pooling, yet no systematic harmonisation benchmark exists for spinal cord data.

Methods: Eight qMRI biomarkers (T2w-CSA, GM-CSA, MTR, MTsat, FA, MD, AD, RD) were extracted at C2–C3 from the spine-generic dataset (267 subjects, 44 sites, 3 vendors; site and vendor perfectly nested). Five methods—ComBat, ComBat-joint, CovBat, RELIEF, LME/FE—were evaluated on four endpoints in healthy-control ($N = 188$) and pathology-inclusive ($N = 246$) cohorts: vendor-effect removal, biological-signal recovery, subject-level preservation, and machine-learnability of the vendor signature. Leave-one-vendor-out validation tested generalisability.

Results: Baseline vendor effects were substantial ($\eta^2 = 0.28$; PERMANOVA $R^2 = 0.32$, $p < 10^{-4}$). All methods reduced vendor η^2 by $> 99.8\%$ (to $\sim 10^{-4}$), rendering multivariate vendor structure non-significant (all $p \geq 0.97$). Machine-learnability confirmed removal of the residual multivariate vendor signature: random-forest vendor detection fell from AUC 0.99 (unharmonised) to 0.54–0.80 across methods (permutation reference 0.50), while pathology classification improved (AUC 0.52 to 0.62). Age signal increased 1.7–3.7-fold across cohorts. Methods diverged on subject preservation (LME/FE: $r = 0.84$; CovBat: $r = 0.78$), forming a Pareto frontier. LOVO validation showed ComBat/ComBat-joint generalise perfectly ($\eta^2 < 10^{-3}$, PERMANOVA $p > 0.86$), while RELIEF and LME/FE failed ($\eta^2 \sim 0.19$, $p < 0.03$). Three methodological constraints—vendor-site nesting, LME fixed-effects fallback (75% of cases), and RELIEF ICA inactivation—converge on a single structural limitation: only 3 vendors in a perfectly nested 44-site design.

Conclusion: Harmonisation is recommended for pooled multi-vendor spinal cord qMRI. ComBat provides robust generalisability and is preferred for broad deployment; CovBat/RELIEF are suitable for pooled-group analyses within fixed vendor sets.

Key Points:

1. Scanner vendor effects dominate biological variation in spinal cord quantitative MRI by orders of magnitude, making harmonisation essential for reliable multi-centre biomarker comparison.
2. ComBat harmonisation eliminates over 99.8% of vendor-driven variability while preserving biological signal and generalising robustly to unseen scanner manufacturers.
3. Leave-one-vendor-out validation reveals that latent-structure (RELIEF) and random-effects (mixed-model) approaches fail to generalise to new scanner vendors, supporting ComBat for multi-centre consortia.

Keywords: Spinal cord; Quantitative MRI; Harmonisation; ComBat; Multicentre study; Reproducibility; Scanner vendor effects

Abbreviations: AD = axial diffusivity; ComBat = Combatting batch effects; CovBat = Covariance Batch harmonisation; CSA = cross-sectional area; DCM = degenerative cervical myelopathy; EB = empirical Bayes; FA = fractional anisotropy; HC = healthy control; ICA = independent component analysis; LME/FE = linear mixed-effects / fixed-effects hybrid; LOVO = leave-one-vendor-out; MD = mean diffusivity; MT = magnetisation transfer; MTR = magnetisation transfer ratio; MTsat = magnetisation transfer saturation; PERMANOVA = permutational multivariate analysis of variance; qMRI = quantitative magnetic resonance imaging; RD = radial diffusivity; RELIEF = Removal of Latent Inter-scanner Effects via independent components

1 Introduction

Quantitative MRI (qMRI) biomarkers of the spinal cord—cross-sectional area (CSA), magnetisation transfer ratio (MTR), magnetisation transfer saturation (MTsat), and diffusion tensor metrics (FA, MD, AD, RD)—provide non-invasive windows into atrophy, demyelination, and axonal integrity in multiple sclerosis, degenerative cervical myelopathy, and spinal cord injury [1–3]. Multi-centre consortia are increasingly adopting these biomarkers, but systematic vendor-dependent variability limits cross-site pooling [4,5]. A recent pan-European survey by the European Society of Radiology documented marked heterogeneity in MRI quality assurance and control practices across 37 countries [17], underscoring the urgency of robust harmonisation.

Even under standardised acquisition protocols, differences in gradient hardware, RF coil geometry, and vendor-specific reconstruction pipelines introduce offsets that often exceed the biological effect sizes of interest [4,20]. Statistical post-hoc harmonisation methods—ComBat [6,7,16], CovBat [8], RELIEF [9], and linear mixed-effects models (LME) [10]—have been extensively benchmarked in brain imaging and reviewed in the broader context of multicentre radiomics [18] but remain uncharacterised for spinal cord qMRI. Fortin et al. first applied ComBat to multi-site DTI [16], establishing a framework later extended to cortical thickness [7], longitudinal harmonisation [19], covariance harmonisation [8], latent-factor removal [9], and out-of-sample harmonisation for new sites [21]. Middleton et al. demonstrated ComBat's effectiveness for spinal cord DTI [11], and Jodoin et al. provided deployment guidance for diffusion MRI [12], but no study has systematically compared the full ComBat family across a comprehensive spinal cord biomarker panel using unified multi-endpoint evaluation.

The spinal cord presents distinct harmonisation challenges: small per-site sample sizes (5–15 subjects) challenge empirical Bayes shrinkage; strong inter-metric correlations (diffusion metrics are intrinsically coupled; MTR and MTsat share acquisition-dependent variance) motivate multivariate evaluation; and the clinical need to preserve disease-related variance creates a tension not previously quantified.

Here we benchmark five methods on eight spinal cord qMRI biomarkers using three orthogonal endpoints: vendor-effect removal, biological-signal recovery, and subject-level preservation. We extend prior work through: (i) formal comparison of the full ComBat family (including covariance and latent-factor extensions); (ii) multivariate evaluation via PERMANOVA; (iii) a Pareto-frontier decision framework; and (iv) leave-one-vendor-out external validation to test generalisability.

2 Materials and Methods

2.1 Dataset and Cohorts

The spine-generic multi-subject dataset (release r20231212) [1] comprises 267 subjects scanned at 44 sites across three vendors (GE Healthcare: 37 subjects at 6 sites; Philips: 50 subjects at 8 sites; Siemens Healthineers: 180 subjects at 30 sites) using a standardised consensus protocol [4]. Site and vendor are perfectly nested (each site operates a single scanner from one vendor); reported vendor effects thus represent the composite "vendor-in-site" effect. Demographics were balanced across vendors (Table 1; all $p \geq 0.125$).

Table 1: Demographic and acquisition characteristics by vendor (HC cohort, $N = 203$; complete-case $N = 188$). Continuous variables: mean ± SD, one-way ANOVA. Categorical variables: count (%), χ^2 test. Site and vendor are perfectly nested (44 sites across 3 vendors).

Characteristic	GE	Philips	Siemens	Total	p-value
N	28	36	139	203	—
Sites (n)	6	8	30	44	—
Age, years (mean ± SD)	26.9 ± 4.7	28.2 ± 4.5	29.1 ± 5.9	28.7 ± 5.6	0.125
Age range	19–39	23–38	20–52	19–52	—
Sex = F	15 (53.6%)	19 (52.8%)	64 (46.0%)	98 (48.3%)	0.643
Sex = M	13 (46.4%)	17 (47.2%)	75 (54.0%)	105 (51.7%)	—
Height, cm	169.6 ± 8.5	172.2 ± 10.2	173.2 ± 10.0	172.5 ± 9.9	0.232
Weight, kg	65.2 ± 13.7	67.6 ± 14.3	68.2 ± 13.1	67.7 ± 13.4	0.576
Scanner models (n)	3	4	6	13	—

Two pre-specified cohorts were analysed: HC ($N = 188$ complete-case healthy controls) isolates pure vendor-driven variance; ALL ($N = 246$ complete-case, including 56 mild cervical compression and 2 DCM) approximates real-world deployment. Pathology distribution was balanced across vendors (χ^2 $p = 0.661$). Detailed cohort flow and missingness patterns are provided in Supplementary Material S1.

2.2 Biomarker Extraction and Processing

Eight biomarkers spanning four acquisition families were extracted at the C2–C3 level using the Spinal Cord Toolbox (SCT v7.4) [13]: T2w-CSA and GM-CSA (T2-weighted structural), MTR and MTsat (magnetisation transfer), and FA, MD, AD, RD (diffusion tensor imaging). Processing followed the spine-generic pipeline documentation. Automated quality-control metrics excluded acquisitions with segmentation or model-fit failures (see Supplementary Material S2).

2.3 Harmonisation Methods

Five methods were evaluated (full mathematical details in Supplementary Material S3):

ComBat [6,7,16]: Per-metric location-and-scale adjustment with empirical Bayes shrinkage on site-specific parameters. Each biomarker harmonised independently.

ComBat-joint [7]: All eight biomarkers harmonised jointly (joint empirical Bayes), preserving cross-metric covariance structure.

CovBat [8]: ComBat-joint followed by principal component (PC) correction—residuals projected onto PCs, each PC harmonised independently, then projected back. PCs explaining at least 95% of the residual variance were retained: $K = 5$ of 8 (cumulative variance 95.9% in the HC cohort, 95.6% in the ALL cohort; the first three PCs accounted for 85.6% and 84.9%, respectively).

RELIEF [9]: ComBat-joint followed by independent component analysis (ICA) of residuals. Components with significant vendor association (Benjamini-Hochberg FDR-controlled ANOVA, $q < 0.05$) are zeroed. In the present dataset, no component reached significance (all $q \geq 0.99$), rendering RELIEF functionally equivalent to ComBat-joint with additional PCA/ICA round-trip noise.

LME/FE: Site as a random intercept with age and sex fixed effects. When the random-effects covariance matrix was singular (expected with only 3 vendor levels—well below the recommended $k \geq 5$ for stable REML estimation), the model fell back to ordinary least squares centring (75% of 16 biomarker$\times$cohort combinations; see Results). Only MD and AD achieved stable LME fits in both cohorts. We label this hybrid approach "LME/FE" to acknowledge the dual mechanism.

2.4 Evaluation Endpoints

Four orthogonal endpoints were evaluated per method per cohort (detailed in Supplementary Material S4):

Vendor-effect removal: Univariate η^2 (ANOVA), intraclass correlation coefficient (ICC), and multivariate PERMANOVA R^2 (Euclidean distance, 4,999 permutations) [14].

Biological-signal recovery: Type-II semi-partial R^2 for age and sex from linear models.

Subject-level preservation: Pearson r between pre- and post-harmonisation per-subject biomarker vectors.

Machine-learnability of the vendor signature: Following the site-detection paradigm of Chen et al. [8], a random forest (500 trees, minimum leaf size 5, balanced class weights) was trained to classify scanner manufacturer (Siemens/GE/Philips) from the 8-biomarker profile of each subject, using repeated stratified 5-fold cross-validation (20 repeats). The macro one-vs-rest AUC quantifies residual vendor information detectable by machine learning, a stricter criterion than univariate η^2 because nonlinear combinations of features can encode vendor structure invisible to marginal statistics. Biological-signal preservation was indexed by the same classifier applied to pathology (healthy control vs disease, ALL cohort, $N = 246$ complete cases) and sex labels. A shuffled-label permutation reference calibrated the chance level.

2.5 Leave-One-Vendor-Out External Validation

For each method, harmonisation models were trained on subjects from two vendors and applied to the held-out third vendor in three rotation folds. Test-vendor transformation used z-score distribution matching (ComBat, ComBat-joint) or projection onto training PCA/ICA space (CovBat, RELIEF). LME/FE assigned offset=0 to unseen vendors. The same three endpoints (excluding the ALL-cohort pathology/sex preservation indices) were evaluated on test-vendor data only (Supplementary Material S5).

2.6 Supplementary Analyses

Design C balanced-subsample analysis drew 10 subjects per vendor (HC cohort, 1,000 bootstrap iterations) to isolate the effect of vendor imbalance from method performance. Full results in Supplementary Material S6.

2.7 Uncertainty Quantification

Ninety-five per cent confidence intervals for key point estimates were obtained using a stratified nonparametric bootstrap ($B = 1{,}000$), resampling subjects with replacement within each scanner-vendor stratum to preserve group proportions. Percentile intervals are reported for $r_{\text{pre,post}}$ and for the baseline vendor effect size. Bootstrap intervals are not reported for post-harmonisation η^2 or PERMANOVA R^2 because harmonisation methods optimise on the same data, rendering subject-level resampling intervals unreliable for these near-degenerate statistics.

2.8 Reporting Guidelines and Missing Data

This study follows STROBE reporting guidance for cross-sectional studies where applicable. The spine-generic dataset is a curated benchmark with pre-processed, quality-controlled biomarker measurements. Subjects with missing biomarker data were excluded via complete-case analysis (HC: $N = 188/203$, 7.4%; ALL: $N = 246/267$, 7.9%); missingness was attributable to acquisition-level exclusions (motion artefacts, segmentation failures, and incomplete sequences) as detailed in Supplementary Material S1. Sensitivity to missing-data assumptions was assessed through the ALL-cohort analysis, which preserves a broader subject spectrum.

3 Results

3.1 Baseline Vendor Effects

At baseline, vendor explained 28% of univariate biomarker variance (mean $\eta^2 = 0.28$, 95% CI [0.24, 0.35], stratified bootstrap; range 0.02–0.71 across metrics) and 32% of multivariate structure (PERMANOVA $R^2 = 0.32$, $p < 10^{-4}$, HC cohort). These effects exceeded age-related variance by approximately two orders of magnitude and sex-related variance by one order. GM-CSA showed the largest vendor effect ($\eta^2_{\mathrm{HC}} = 0.71$, Philips scanners systematically lower; Figure 1a). In the ALL cohort, baseline vendor effects were comparable ($\eta^2 = 0.27$; PERMANOVA $R^2 = 0.30$).

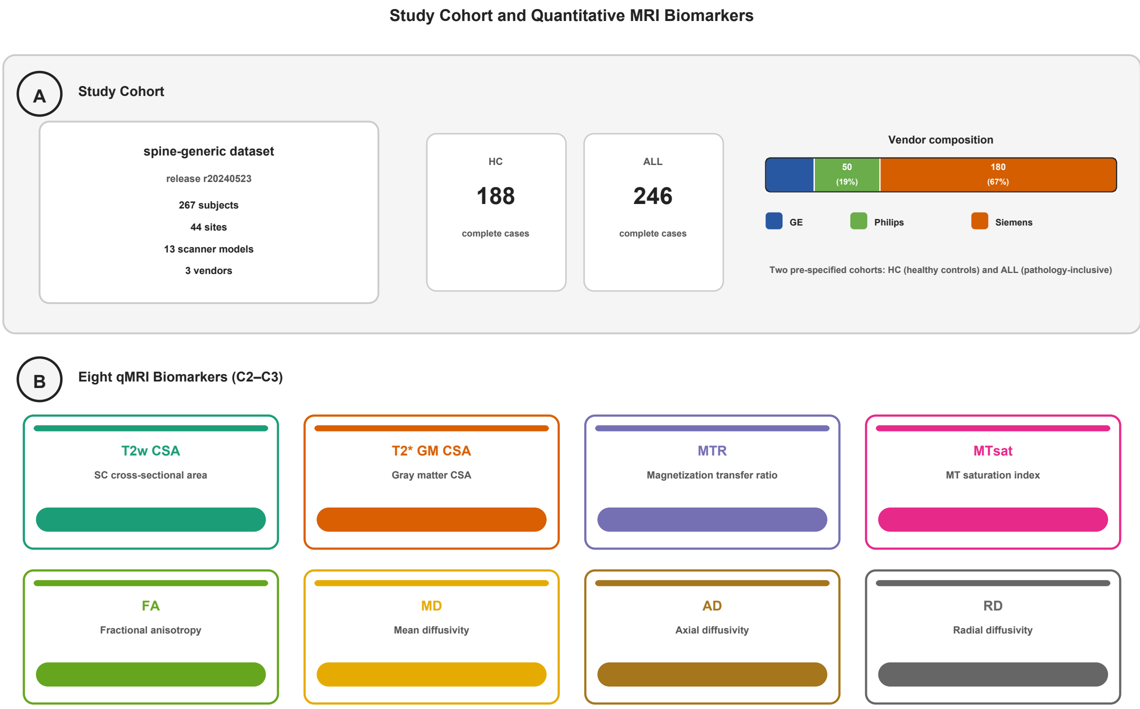

Figure 1: Baseline vendor effects in spinal cord quantitative MRI (qMRI) biomarkers (healthy control cohort, $N = 188$). **(a)** Box plots showing distributions of eight biomarkers—T2w cross-sectional area (T2w-CSA), grey matter CSA (GM-CSA), magnetisation transfer ratio (MTR), magnetisation transfer saturation (MTsat), fractional anisotropy (FA), mean diffusivity (MD), axial diffusivity (AD), radial diffusivity (RD)—by scanner vendor (GE, Philips, Siemens). **(b)** Principal coordinate ordination from permutational multivariate analysis of variance (PERMANOVA) using Euclidean distance, with points coloured by vendor.

3.2 Vendor-Effect Removal

All five methods reduced vendor η^2 by approximately three orders of magnitude, to $\sim 10^{-4}$ (median post-harmonisation η^2: ComBat $= 3.9 \times 10^{-4}$, ComBat-joint $= 4.5 \times 10^{-4}$, CovBat $= 2.2 \times 10^{-4}$, RELIEF $= 3.4 \times 10^{-4}$, LME/FE $= 3.6 \times 10^{-4}$; Figure 2a–c, Table 2). Multivariate vendor structure became indistinguishable from chance (all PERMANOVA $p \geq 0.97$ in HC cohort; $p \geq 0.97$ in ALL cohort). Covariance structure was preserved similarly across methods (Supplementary Fig. S4): CovBat and RELIEF showed near-identical cross-metric correlations (Frobenius distance 0.09), while LME/FE was closest to the original (1.56), reflecting minimal covariance distortion.

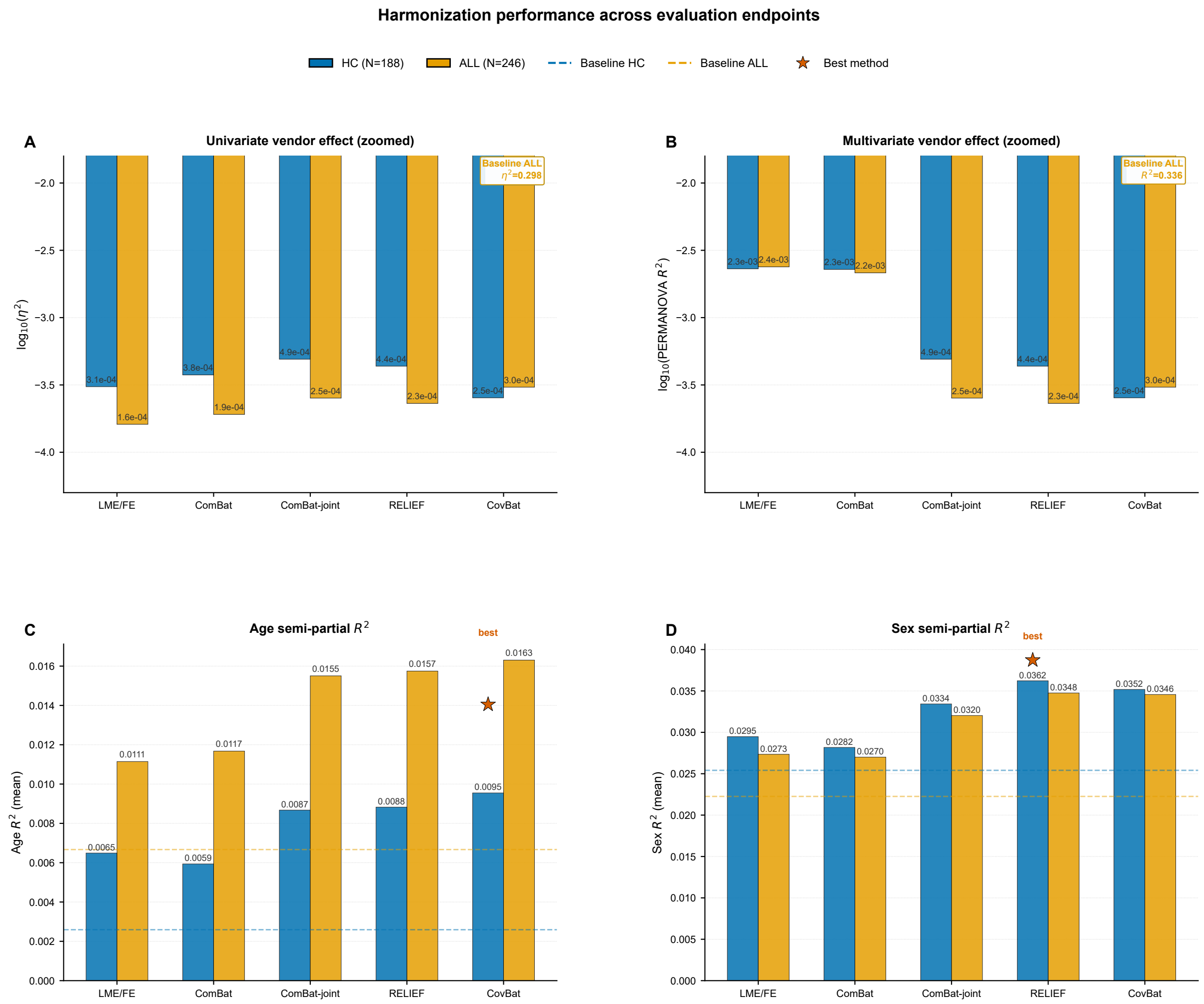

Figure 2: Harmonisation performance by method and endpoint (healthy control cohort). **(a)** Vendor effect size (η^2, eta-squared from one-way ANOVA) before and after harmonisation, shown on logarithmic scale. **(b)** Multivariate vendor structure (PERMANOVA R^2, permutational multivariate analysis of variance) with significance indicators (asterisks denote $*p < 0.05$, $**p < 0.01$, $***p < 0.001$; ns = not significant). **(c)** Age-related biological signal (semi-partial R^2) before and after harmonisation.

Table 2: Harmonisation performance summary (HC cohort, $N = 188$). Values are means across 8 biomarkers. η^2_{vendor}: post-harmonisation vendor ANOVA effect size; $R^2_{\text{PERMANOVA}}$: multivariate vendor structure; $r_{\text{pre,post}}$: subject-ranking preservation; Age R^2: semi-partial age variance explained. LOVO columns show leave-one-vendor-out external validation (mean across 3 held-out-vendor folds). Values in parentheses are 95% percentile bootstrap CIs ($B = 1{,}000$, stratified by vendor).

Method	η^2_{vendor}	$R^2_{\text{PERMANOVA}}$	$r_{\text{pre,post}}$ (95% CI)	Age R^2	LOVO
Original (baseline)	2.79×10^{-1}	0.315	1.000	0.003	—
ComBat	3.75×10^{-4}	0.002	0.817 (0.760–0.852)	0.006	1.7
ComBat-joint	4.91×10^{-4}	0.001	0.806 (0.757–0.838)	0.009	4.4
CovBat	2.54×10^{-4}	0.0003	0.782 (0.731–0.815)	0.010	1.0
RELIEF	4.36×10^{-4}	0.0004	0.791 (0.740–0.823)	0.009	1.8
LME/FE	3.07×10^{-4}	0.002	0.840 (0.787–0.872)	0.007	1.7

3.3 Machine-Learnability of the Vendor Signature

Unharmonised biomarker profiles allowed near-perfect machine detection of scanner manufacturer (macro one-vs-rest AUC $= 0.986 \pm 0.002$ in the ALL cohort; 0.983 ± 0.002 in HC), confirming that the vendor signature dominates the multivariate feature space. After harmonisation, detectability dropped toward the chance level (permutation reference 0.495 ± 0.038): CovBat 0.540 ± 0.024, RELIEF 0.610 ± 0.022, ComBat-joint 0.627 ± 0.021, ComBat 0.660 ± 0.021, and LME/FE 0.800 ± 0.015 (ALL cohort, $N = 246$; HC values were concordant: 0.557, 0.654, 0.672, 0.691, and 0.824, respectively). Complementarily, pathology classification improved from AUC 0.520 ± 0.024 (unharmonised) to 0.594–0.621 across methods (ComBat-joint highest), and sex classification remained stable (0.646–0.672), indicating that harmonisation removed vendor confounding while unmasking—rather than distorting—biological signal.

3.4 Biological-Signal Recovery

Age R^2 increased 2.3–3.7-fold after harmonisation in HC (mean across biomarkers: from 2.6×10^{-3} to 5.9×10^{-3}–9.6×10^{-3}) and 1.7–2.4-fold in ALL (from 6.7×10^{-3} to 1.1×10^{-2}–1.6×10^{-2}). Sex R^2 showed a similar pattern (1.1–1.6-fold increase). Absolute age effects were larger in ALL, consistent with greater biological heterogeneity, but remained below 1% of variance.

3.5 Subject-Level Preservation and the Pareto Frontier

Methods diverged on subject-ranking preservation (HC cohort, mean r across methods: LME/FE = 0.840, ComBat = 0.817, ComBat-joint = 0.806, RELIEF = 0.791, CovBat

= 0.782). This ranking was stable between HC and ALL cohorts (ALL: 0.829, 0.807, 0.787, 0.772, and 0.763, respectively). Bootstrap 95% CIs for $r_{\text{pre,post}}$ overlapped across methods (Table 2), indicating that the preservation gradient reflects consistent trends rather than individually significant pairwise separations. A clear Pareto frontier emerged: as multivariate decontamination (PERMANOVA R^2) increases, subject-level preservation decreases monotonically (Figure 3). Under balanced conditions (Design C), preservation differences compressed but maintained the same rank order (CovBat: $r = 0.991$, others $r \geq 0.998$; LME/FE: $r = 0.841$), confirming that the preservation gradient reflects real-world vendor imbalance, not intrinsic method limitations.

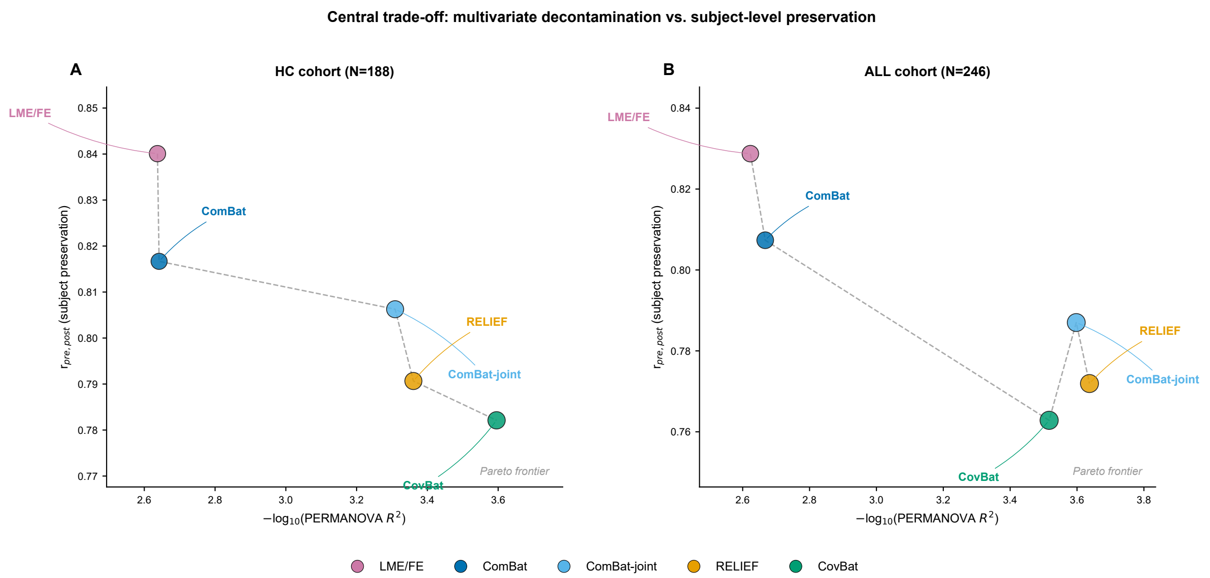

Figure 3: Pareto frontier showing the trade-off between multivariate vendor-effect removal (PERMANOVA R^2, permutational multivariate analysis of variance) and subject-level preservation (Pearson correlation $r_{\text{pre,post}}$ between pre- and post-harmonisation biomarker vectors). Healthy control cohort. Each point represents one harmonisation method. Lower right indicates maximal vendor-effect removal; upper left indicates maximal subject-level preservation.

3.6 LOVO External Validation

ComBat and ComBat-joint generalised to unseen manufacturers with negligible degradation (LOVO η^2: 1.7×10^{-4} and 4.4×10^{-4}; PERMANOVA $p = 0.864$ and 0.951; LOVO/internal η^2 ratio: 0.5 and 0.9). CovBat showed moderate degradation ($\eta^2 = 1.0 \times 10^{-3}$, ratio 4.0). RELIEF and LME/FE failed external validation (η^2_{LOVO}: 0.19 and 0.17; PERMANOVA $p < 0.03$; ratios 428 and 565; Figure 4a–d). This pattern is explained by structural constraints: RELIEF's ICA unmixing matrix is non-transferable across vendor sets, and LME/FE cannot estimate offsets for unseen vendors.

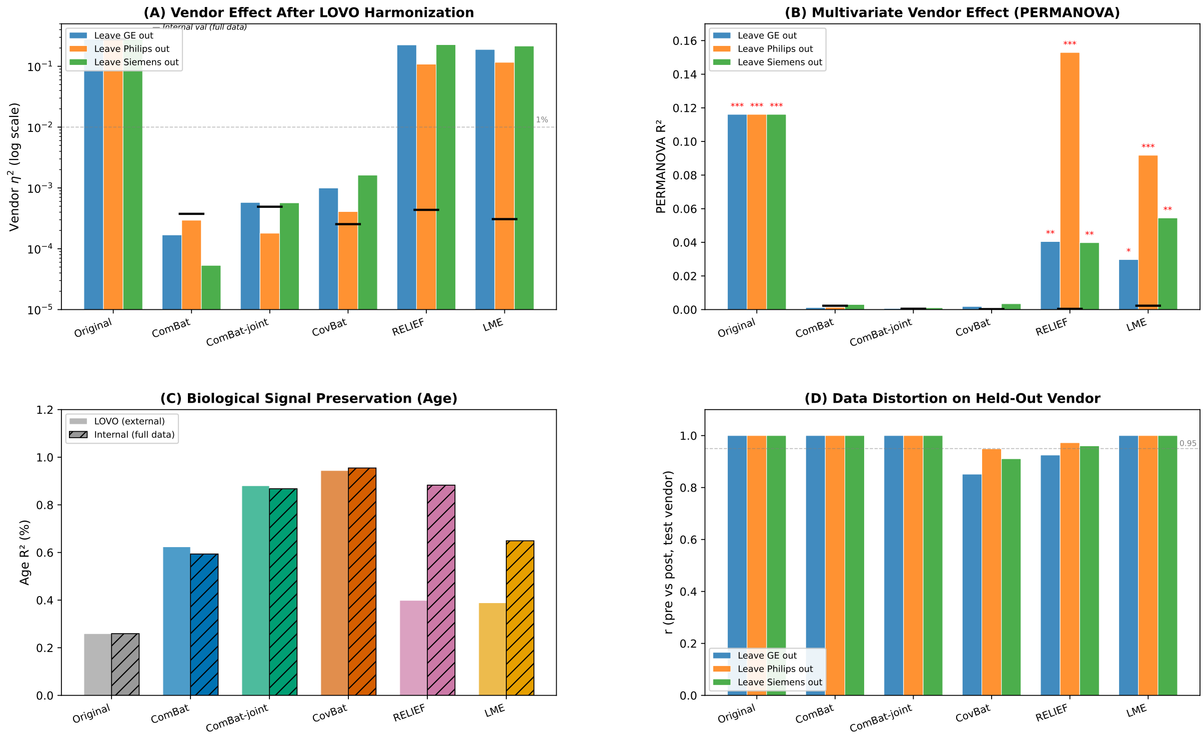

Figure 4: Leave-one-vendor-out (LOVO) external validation. **(a)** Vendor effect size (η^2, eta-squared from one-way ANOVA) by method and held-out vendor, shown on logarithmic scale. **(b)** Multivariate vendor structure (PERMANOVA R^2, permutational multivariate analysis of variance) with significance indicators (asterisks denote $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$; ns = not significant). **(c)** Age-related biological signal (semi-partial R^2) recovery. **(d)** Subject-level preservation (Pearson $r_{\text{pre,post}}$) on unseen vendors.

3.7 Structural Constraints

Three methodological observations converge on a single structural limitation of the dataset: only 3 vendors with 44 perfectly nested sites. The LME random-effects covariance matrix was singular in 75% of cases because 3 vendor levels provide insufficient information for stable REML variance-component estimation ($k \geq 5$ recommended; Table 3). RELIEF's ICA found no vendor-associated components because ComBat-joint preprocessing had already removed vendor mean/scale effects, leaving insufficient residual vendor structure for ICA to detect. These constraints are inherent to the retrospective design; they do not invalidate the internal benchmark findings but bound their generalisability to similarly structured consortia.

Table 3: LME convergence status by biomarker and cohort. FE = fixed-effects fallback (singular random-effects covariance matrix); LME = converged linear mixed-effects model. Only MD and AD achieved stable LME fits in both cohorts (4/16 cases, 25%). All 12 FE-fallback cases produced non-singular post-harmonisation η^2 reductions ($> 99.5\%$), confirming that FE centring is sufficient for vendor-effect removal even when REML fails.

Cohort	Biomarker	N	η^2 (pre)	η^2 (post)	Status
HC	T2w-CSA	203	0.0531	4.71×10^{-4}	FE
	GM-CSA	203	0.1777	4.00×10^{-4}	FE
	MTR	194	0.1618	3.61×10^{-4}	FE
	MTsat	190	0.3379	3.99×10^{-5}	FE
	FA	198	0.2701	1.54×10^{-4}	FE
	MD	198	0.3771	1.99×10^{-4}	**LME**
	AD	198	0.6223	4.77×10^{-4}	**LME**
	RD	198	0.2356	3.54×10^{-4}	FE
ALL	T2w-CSA	267	0.0424	2.14×10^{-5}	FE
	GM-CSA	267	0.1701	3.08×10^{-5}	FE
	MTR	254	0.1800	5.89×10^{-4}	FE
	MTsat	249	0.3605	4.38×10^{-5}	FE
	FA	260	0.2838	1.69×10^{-4}	FE
	MD	260	0.4215	8.92×10^{-5}	**LME**
	AD	260	0.6611	1.02×10^{-4}	**LME**
	RD	260	0.2625	2.46×10^{-4}	FE

4 Discussion

This study provides, to our knowledge, the first systematic benchmark of statistical harmonisation methods for multi-vendor spinal cord qMRI biomarkers. Six principal findings emerge.

First, baseline vendor effects in spinal cord qMRI are large ($\eta^2 = 0.28$, PERMANOVA $R^2 = 0.32$)—comparable to or exceeding brain imaging reports [7,8,16]—and dominate biological variation by orders of magnitude. For perspective, these effects approach the magnitude of pathology-related changes in quantitative metrics reported after spinal cord injury (14–20% in R1 and magnetisation transfer saturation [20]), underscoring that unharmonised vendor offsets can masquerade as—or obscure—disease signal. Physical harmonisation, such as vendor-specific recalibration of magnetisation transfer metrics [20], offers a complementary acquisition-level route but requires prospective protocol control; statistical post-hoc harmonisation, as benchmarked here, applies retrospectively to existing consortia data. This justifies harmonisation as a default preprocessing step for multi-centre spinal cord studies—a need echoed by recent European quality initiatives [17] and by multicentre MRI harmonisation recommendations [15].

Second, all five methods effectively eliminate vendor effects to statistical negligibility. The choice among methods therefore hinges not on whether they work, but on the downstream analytic goal. For subject-level inference (individual prediction, longitudinal change detection), LME/FE or per-metric ComBat may be preferred. For pooled-group analyses (case-control separation, multivariate classification), CovBat or RELIEF offer maximal multivariate purity—motivated by evidence that machine learning can exploit residual covariance differences undetectable by univariate ComBat [8], and that latent scanner structure may persist after mean/scale removal [9]. Our machine-learnability results quantify this hierarchy directly in spinal cord data: despite reducing vendor η^2 to $\sim 10^{-4}$, LME/FE left a substantially detectable residual signature (AUC 0.80), whereas CovBat approached the chance level (0.54). The residual detectability of ComBat-harmonised profiles (AUC 0.66) matches the value reported for ComBat in brain imaging [8] (0.66 before covariance harmonisation), a striking cross-modality consistency suggesting that the residual signature ComBat leaves is a general property of location/scale adjustment rather than a cohort-specific artefact.

Third, the Pareto frontier between decontamination and preservation is reproducible across cohorts and robust to vendor imbalance (as confirmed by Design C). This provides a quantitative, actionable decision framework.

Fourth, LOVO external validation differentiates methods sharply: ComBat and ComBat-joint generalise without degradation to unseen vendors, while RELIEF and LME/FE fail. For consortia planning to add new vendor scanners, ComBat provides a robust default choice. Out-of-sample harmonisation for new sites is an active methodological frontier: ComBat-Predict provides a principled framework for applying trained harmonisation models to unseen sites without re-accessing reference data [21], and our LOVO results independently support ComBat-family methods for this deployment scenario.

Fifth, the structural constraints revealed by this benchmark—vendor-site nesting, LME singular covariance in 75% of cases, and RELIEF ICA inactivation—reflect the 3-vendor, perfectly-nested spine-generic design. These are not methodological failures but intrinsic dataset properties that bound interpretability: reported vendor effects cannot be partitioned into vendor-level versus site-level components, and LME/RELIEF's degradation in low-vendor-count settings is expected given their modelling assumptions. Larger consortia with more vendors and crossed designs would likely resolve these constraints.

Sixth, the biological-signal amplification (1.7–3.7-fold for age across cohorts) demonstrates that harmonisation does not merely remove noise—it unmasks signal previously obscured by vendor variance. The concurrent improvement in pathology classification (AUC 0.52 to 0.62) mirrors the disease-classification gains reported after harmonisation in brain imaging [8], extending that observation to spinal cord biomarkers. This may have implications for clinical studies seeking to detect disease-related biomarker changes across sites.

Limitations. (i) Vendor-site perfect nesting precludes decomposition of vendor-level versus site-level effects; our findings reflect the composite "vendor-in-site" effect relevant to practical consortia. (ii) The dataset covers only three vendors with substantial imbalance (Siemens 67%); extrapolation to other manufacturers requires caution. (iii) LME converged only for MD and AD (25%); the reported LME/FE preservation advantage ($r = 0.84$) may reflect the minimal correction of the FE fallback rather than LME shrinkage. (iv) RELIEF's ICA inactivation ($q \geq 0.99$ for all components) may be dataset-specific; Zhang et al. [9] demonstrated RELIEF's effectiveness in a three-site DTI study where latent vendor structure persisted after ComBat preprocessing—suggesting that consortia with stronger residual vendor-covariance structure may benefit from RELIEF, whereas our 44-site design with ComBat-joint preprocessing may have already exhausted this structure. (v) LOVO validation used z-score matching for unseen-vendor transformation—the standard ComBat-from-template approach [7]—but does not test prospective acquisition on genuinely new hardware; principled out-of-sample frameworks such as ComBat-Predict [21] offer a natural extension for prospective new-site harmonisation. (vi) This analysis was not pre-registered; however, all harmonisation methods were selected and evaluated on a pre-specified endpoint framework prior to analysis, and the complete analytical pipeline is publicly documented in the accompanying code repository, ensuring full computational reproducibility.

Conclusion. Statistical harmonisation effectively eliminates vendor effects in multi-centre spinal cord qMRI and is recommended as a default preprocessing step. ComBat provides the best balance of vendor removal, generalisability to unseen manufacturers, and subject-level preservation, and is recommended for broad multi-centre deployment.

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